```
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn
```

```
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.5.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->d
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hu
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
```

Start coding or generate with AI.

```
import pandas as pd
import torch
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
from transformers import DistilBertTokenizerFast
from transformers import DistilBertForSequenceClassification
from transformers import Trainer, TrainingArguments
```

```
data = pd.read_csv("sentiment_dataset.csv")

print(data.head())
```

```
                              text  label
0                I love this phone      1
1          This product is amazing      1
2               Worst service ever      0
3  I am very happy with the results      1
4             This app is terrible      0
```

Start coding or generate with AI.

```
data = data.dropna()

texts = data['text'].tolist()
labels = data['label'].tolist()
```

Start coding or generate with AI.

```
train_texts, test_texts, train_labels, test_labels = train_test_split(
    texts,
    labels,
    test_size=0.2,
    random_state=42
)
```

Start coding or generate with AI.

```
tokenizer = DistilBertTokenizerFast.from_pretrained('distilbert-base-uncased')

train_encodings = tokenizer(train_texts, truncation=True, padding=True)
test_encodings = tokenizer(test_texts, truncation=True, padding=True)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 1.97kB/s]
vocab.txt: 232k/? [00:00<00:00, 4.25MB/s]
tokenizer.json: 466k/? [00:00<00:00, 5.75MB/s]
```

Start coding or generate with AI.

```
class SentimentDataset(torch.utils.data.Dataset):

    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels

    def __getitem__(self, idx):

        item = {key: torch.tensor(val[idx]) for key, val in self.encodings.items()}
        item['labels'] = torch.tensor(self.labels[idx])

        return item

    def __len__(self):
        return len(self.labels)
```

Start coding or generate with AI.

```
train_dataset = SentimentDataset(train_encodings, train_labels)
test_dataset = SentimentDataset(test_encodings, test_labels)
```

Start coding or generate with AI.

```
model = DistilBertForSequenceClassification.from_pretrained(
    'distilbert-base-uncased',
    num_labels=2
)
```

```
config.json: 100%                                                    483/483 [00:00<00:00, 1.86kB/s]

model.safetensors: 100%                                              268M/268M [00:02<00:00, 172MB/s]

Loading weights: 100%                          100/100 [00:00<00:00, 442.30it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_transform.bias     | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
pre_classifier.weight    | MISSING    |
classifier.bias          | MISSING    |
classifier.weight        | MISSING    |
pre_classifier.bias      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

Start coding or generate with AI.

```
training_args = TrainingArguments(

    output_dir="./results",

    num_train_epochs=3,

    per_device_train_batch_size=8,

    per_device_eval_batch_size=8,

    eval_strategy="epoch",

    logging_dir="./logs",

    logging_steps=10
)
```

```
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.
```

Start coding or generate with AI.

```
trainer = Trainer(

    model=model,

    args=training_args,

    train_dataset=train_dataset,

    eval_dataset=test_dataset

)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as tru
  super().__init__(loader)
```

[15/15 00:25, Epoch 3/3]

Epoch	Training Loss	Validation Loss
1	No log	0.674555
2	0.649990	0.608357
3	0.649990	0.558254

```
Writing model shards: 100%                                          1/1 [00:00<00:00, 1.47it/s]
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as tru
  super().__init__(loader)
TrainOutput(global_step=15, training_loss=0.6020821412404378, metrics={'train_runtime': 29.4223,
'train_samples_per_second': 4.079, 'train_steps_per_second': 0.51, 'total_flos': 279423419040.0, 'train_loss':
0.6020821412404378, 'epoch': 3.0})
```

Start coding or generate with AI.

```
predictions = trainer.predict(test_dataset)

preds = predictions.predictions.argmax(axis=1)

accuracy = accuracy_score(test_labels, preds)

print("Accuracy:", accuracy)

print(classification_report(test_labels, preds))
```

```
Accuracy: 0.9
              precision    recall  f1-score   support

           0       1.00      0.83      0.91         6
           1       0.80      1.00      0.89         4

    accuracy                           0.90        10
   macro avg       0.90      0.92      0.90        10
weighted avg       0.92      0.90      0.90        10
```